EXTENSIVE README_TG REPOSITORY

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Description

EVAP-COND and ACSIM use refrigerant property look-up tables to enable fast simulations. These tables contain the full set of thermodynamic and transport properties. The look-up scheme includes eight property routines, each designed to retrieve a specific state or transport property based on the variables defining the refrigerant's thermodynamic state. The tables are formulated in pressure–enthalpy coordinates and can extend into the supercritical region for high-pressure refrigerants. The look-up tables distributed with EVAP-COND and ACSIM were generated using TableGen2.exe, a stand-alone program based on the property routines in REFPROP [1].

The installation module for TableGen2.exe is available on the NIST ACSIM and EVAP-COND pages (https://www.nist.gov/services-resources/software/acsim and https://www.nist.gov/services-resources/software/evap-cond-version-50) and directly within this repository. This repository also provides complete source code for TableGen2.exe except for the proprietary REFPROP subroutines for calculating refrigerant properties, which can be obtained from REFPROP. In either case, the user also needs to obtain REFPROP-compatible 'fluids files' for refrigerants of interest.

Repository Contents

FORTRAN_TG_Files:
All source code files for the table generator, TableGen2exe, except for the proprietary REFPROP source code files (to be obtained from [1]).

FORTRAN_VS_TG_Project:
Visual Studio project for the table generator. Does not include the REFPROP source code files. Includes REFPROP fluid files for R-134a and propane only. REFPROP fluid files for other refrigerants of interest must be provided by the user (can be obtained from [1]).

Installer_TG:
Installation module setup files for the table generator.

Documentation:
EXTENSIVE README_TG REPOSITORY.pdf (this document)
TableGen2_Info.pdf

GUI_VS_TG_Project:
Visual Studio project for the GUI

Refer to TableGen2_Info.pdf contained in this repository for instructions on the installation and use of the table generator.

Reference

[1] Lemmon E, Bell IH, Huber M, McLinden M (2018) NIST Standard Reference Database 23: Reference Fluid Thermodynamic and Transport Properties-REFPROP, Version 10.0, National Institute of Standards and Technology. *Standard Reference Data Program, Gaithersburg*, MD. https://www.nist.gov/srd/refprop

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System/Platform Requirements
Personal computer with Windows 10 or Window 11 operating system
Free space for complete installation: approximately 2 MB